Two Pipelines Compared: NxN Super-Embedding Matrix vs. Final Crop-Re-encode + Zoom-Refine

SAM2-only, training-free co-saliency. All numbers measured on the 16-image alarm-clock group (+ CoCA dev subset where noted).

1. What each pipeline is

Matrix pipeline -- the original formulation. One `NxN` block matrix whose cell `(i,j) = E■ . E■^T` is the full block of dot-products between the **club super-embeddings** of image i and image j (diagonal / intra-image blocks skipped). A club's co-saliency score is the consensus over its row of blocks (max or mean over the other N–1 images) minus a background penalty; a selection rule turns scores into a mask.

Final pipeline -- the endpoint. Same SAM2 partition, but each club is compared as a **set of superpixel embeddings** (not a mean), finalists are **re-encoded by SAM2 at crop scale**, selection is a **soft-AND consensus clique** over union candidates, and masks come from **SAM2 decoder zoom refinement**.

The crucial structural difference: the matrix pipeline collapses each club to **one mean vector**; the final pipeline never collapses -- it keeps sets, re-encodes objects, and refines boundaries.

2. Architecture, side by side

Stage	Matrix pipeline	Final pipeline
Per-image partition	SAM2 -> SNIC -> prompts -> proposals -> NMS -> subtraction -> clubs	*(identical)*
Descriptor per club	**mean** super-embedding (256-d, collapsed)	**set** of superpixel embeddings + **SAM2 crop re-encoding** (raw + masked)
Cross-image comparison	`E■ . E■^T` block (mean . mean)	min-Chamfer set.set, then soft-AND `rank(qsim).rank(cos_raw).rank(cos_masked)`
Candidate	single club	club **+ connected unions <= 3**
Selection	argmax / threshold / anchor-grow over club scores	consensus **clique** (multi-restart) + marginal-gain prune + crop-gated expansion
Multiple instances	not modeled	top-3 near-duplicate bar (P25 of winners' top-3)
Mask	partition region (ragged, nearest-upsampled)	**SAM2 decoder zoom refinement**

3. The mask-quality fix (why the final pipeline's masks are "proper")

Three distinct causes were diagnosed; two were fixed by **zoom refinement**.

cause	symptom	status
decoder ran on the whole image -> SAM2's internal 1024 budget diluted across 6000 px	chunky boundaries on HD, thin parts (gear struts, antennas, bell rings) below decode resolution	**fixed** -- decode on the object crop: full 1024 covers just the object (~2.4x sharper on planes, ~6x on small clocks)
prompt box clamped to the seed's bbox	missing bells / handle-tips the club selection never contained	**fixed** -- box padded 60 % beyond the seed so the decoder grows into missing parts
seed itself contains a wrong club	07's blanket patch riding along the clock	**remains** -- this is selection purity, not mask quality; the decoder faithfully segments whatever the seed includes

The matrix pipeline has none of this -- its mask *is* the raw partition region, so it is simultaneously ragged **and** only ever a part (see §4).

4. The matrix pipeline's ceiling, measured

The block matrix scores **individual clubs**, and a club is a *part*, not a whole object. Three selection rules, three behaviours -- all on the identical `E■.E■^T` matrix:

selection rule	result	mask
`argmax` single club	**13/16**	the **bells only** (bells are the most consistent recurring part)

selection rule	result	mask
`threshold + union`	**~11/16**	whole clock **+ bleed** (15 -> 21 bouquet clubs)
`anchor-connected grow`	**~11/16**	less bleed, but 15 still -> 23 clubs; 05/13/14 shrink to 1 club

Two walls no selection rule can pass:

1. **Wrong anchor (appearance ceiling)** -- on 10 (figurine), 11 (teddy), 14 (banana), 15 (bouquet) the club-mean of the distractor genuinely matches clock club-means, so the pick starts on the wrong object.

2. **Threshold fragility** -- no single threshold is right for all 16 images; tight shrinks to a part, loose bleeds.

Independently verified: the NxN block form is **mathematically identical** to the flat `P.P^T` consensus (max abs difference `2e-7`) -- a reorganization, not a new algorithm.

5. How the final pipeline breaks each wall

matrix-pipeline wall	root cause	final-pipeline fix
bells-only answer	scores single clubs (parts)	**connected union candidates** -- whole objects are first-class candidates
wrong anchor on look-alikes	club **mean** embedding can't tell a structured clock from a uniform blob	**min-Chamfer set matching** + **SAM2 crop re-encoding** (object-scale second opinion)
bleed / threshold fragility	free per-club threshold	bounded **clique** selection + marginal-gain prune
ragged / partial masks	raw partition region	**SAM2 decoder zoom refinement** (crop-scale, padded box, LCC + fill)

6. Results, measured

Alarm-clock group (16 images):

pipeline	whole-object correct	mask quality
Matrix (argmax)	13/16	bells only
Matrix (threshold+union)	~11/16	whole + bleed
Final	**16/16**	**whole, decoder-crisp**

Generality (final pipeline, frozen parameters): airplanes (HD Airbus) 16/16 zero-shot, whole airframe, crisp; clover 5/10 strict; dice 2/16 (candidate-recall limited).

CoCA benchmark, 12-group dev subset (final pipeline, pixel GT, standard metrics):

	MAE ↓	Fbeta ↑	Sα ↑	Eξ ↑
Final pipeline	**0.086**	**0.602**	**0.715**	0.773
ZS-CoSOD (prior training-free)	0.115	0.549	0.667	--
TF-SSD (training-free SOTA, CVPR'26)	0.077	0.686	0.763	~0.815
oracle (our candidate ceiling)	--	0.884	--	--

The matrix pipeline was never benchmarked on CoCA -- it plateaus at the ~13/16 appearance ceiling with partial masks, which is the reason it was superseded.

7. Honest bottom line

- The **matrix pipeline** is a correct, clean formulation whose *similarity* is fine -- but it collapses clubs to means and picks single clubs, so it tops out at **13/16 with bells-only (or bleedy) masks**. Its value was diagnostic: it proved that re-organizing the *same* embeddings cannot break the ceiling.

- The **final pipeline** keeps everything the matrix pipeline could not: whole objects (union candidates), a non-collapsed descriptor (set matching + crop re-encoding), bounded selection (clique), and proper boundaries (zoom refinement). Result: **16/16 clean on clocks, 16/16 on HD planes, and second-best-among-training-free on the CoCA dev subset** -- with the remaining gap to SOTA located precisely in *selection under engineered distractors*, not in segmentation.

- One shared, unbroken wall: where a distractor's appearance genuinely matches the target (07 blanket, CoCA cats-with-ducks), both pipelines fail -- the SAM2-only, appearance-based descriptor cannot make a semantic distinction.

That is the honest ceiling of the constraint.